BRUSSELS 2026 HACKATHON / PIZZA4PEOPLE / TECHNICAL

DISCOVERY WITHOUT PLATFORM RELAY.

Use this when the group needs the deeper technical version: discovery can use a registry, but menu and order traffic still go directly from client to node. The registry must not become the order path.

```
announce -> heartbeat -> discover -> menu -> order client -> registry for discovery client
-> node for menu and order backup registry only for failover
```

TECHNICAL CLAIM THE GROUP CAN STAND BEHIND

P4P proves that a local takeaway shop can stay directly reachable without a marketplace middleman owning first contact, while keeping discovery and ordering as separate layers.

CORE WIRE LOOP

1. node announces itself to one or more registries
2. node heartbeats keep stale entries out of discovery
3. client queries registry for nearby nodes
4. client fetches menu from the node endpoint
5. client submits order to the node endpoint
6. client can fail over to a backup registry

```
discover: client -> registry
menu:     client -> node
order:    client -> node
mode:     order_mode = test | live
backup:   client -> backup registry
```

WHY IT IS TECHNICALLY INTERESTING

- registry out of order path
- node-owned endpoint and menu truth
- explicit order consent via `order_mode`
- failover at discovery layer
- modules outside minimum interoperable core

ANTI-CAPTURE BOUNDARY

This is not just another ordering backend. The point is to keep the minimum interoperable core small enough that a shop can remain directly reachable and a registry can help discovery without owning the customer relationship.

TECHNICAL VERSION THE GROUP CAN SHOW

- one small registry
- one small shop node
- one browser client
- one operator screen
- request trace that proves menu/order hit node, not registry
- optional backup registry if time allows

HARD NON-GOALS

- production security story
- online payment rail
- delivery logistics
- restaurant rollout
- mature federation
- trust/certification ecosystem

TECHNICAL MATERIAL ALREADY THERE

- **public concept:** pizza4people.com
- **repo:** github.com/DennisHedegreen/p4p
- **proof boundary:** `PROOF.md`
- **wire contract:** `SPEC.md`
- **wider explainer:** `docs/WHITEPAPER-v0.1.md`